Integration WS 5

Evaluate Q1-15

1. $\int \frac{\ln x}{x} dx$
2. $\int \frac{x+7}{(x-1)(x+3)} dx$
3. $\int \frac{2x^2-2x+1}{(x-2)(x^2+1)} dx$
4. $\int_2^4 \frac{(x^2-1)^2}{x} dx$
5. $\int \frac{dx}{(4+x^2)^{\frac{3}{2}}}$
6. $\int \frac{1}{(1+x)\sqrt{x}} dx$
7. $\int \frac{x}{\sqrt{x+1}} dx \quad (u^2 = x+1)$
8. $\int_{\sqrt{2}}^{2} \frac{1}{x\sqrt{x^2-1}} dx$
9. $\int \sqrt{\sin x} \cos^3 x \, dx$
10. $\int \sin 4x \cos x \, dx$
11. $\int x^2 e^x dx$
12. $\int_0^1 \tan^{-1} x \, dx$
13. $\int_0^2 \frac{5x^2+4x-20}{(x+2)(x^2+4)} dx$
14. $\int_0^{\frac{\pi}{2}} \frac{1}{3\cos x + 4\sin x + 5} dx$
15. $\int_0^a \frac{3x^2-ax}{(x-2a)(x^2+a^2)} dx$
16. a. Show that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

    b. Hence show that $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x} dx = \frac{\pi^2}{4}$
17. a. Show that $\int_0^a f(x)dx = \int_0^a f(a-x)\,dx$

    b. i. Evaluate $\int_0^{\frac{\pi}{4}} \frac{1-\sin 2x}{1+\sin 2x} dx$

    ii. Evaluate $\int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx$
18. a. Show that $\int_{-a}^a f(x)dx = \int_0^a [f(x) + f(-x)]dx$

    b. Hence show that $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^x}{1+e^x} \sin^2 x \, dx = \frac{\pi}{4}$
19. Use the result $\int_{-a}^a f(x)dx = \int_0^a [f(x) + f(-x)]dx$

    a. Evaluate $\int_{-1}^1 \frac{1}{1+e^{-x}} dx$

    b. Evaluate $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1}{1+\sin x} dx$
20. If $I_n = \int_1^e (\ln x)^n \, dx$ for $n \geq 0$, show that $I_n = e - nI_{n-1}$ for $n \geq 1$. Hence evaluate I_4.
21. Show that $f(x) = x^6 \sin x$ is an odd function. Hence evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^6 \sin x \, dx$
22. Show that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$. Hence show that $\int_0^{\frac{\pi}{2}} \frac{\cos x}{\cos x + \sin x} dx = \frac{\pi}{4}$.
23. Find $\int x \sec x \tan x \; dx$
24. Knowing that $(cis\ \theta)^3 = cis\ 3\theta$. Use this result to help determine $\int \cos^3\theta \; d\theta$.
25. Show that $\int_0^{\frac{\pi}{4}} \tan^2 x \sec^3 x \; dx = \frac{1}{8}\left(3\sqrt{2} - \ln\left(\sqrt{2}+1\right)\right)$
26. a. Given that $I_n = \int \tan^n x \, dx$, prove that $I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$ for $n \geq 2$.

    b. Hence show that $I_6 = \frac{1}{5}\tan^5 x - \frac{1}{3}\tan^3 x + \tan x - x + C$
27. a. If $I_n = \int x^n e^x \, dx$, show that $I_n = x^n e^x - nI_{n-1}$ for $n \geq 1$.

    b. Hence show that $\int x^3 e^x \, dx = (x^3 - 3x^2 + 6x - 6)e^x + C$
28. a. If $I_n = \int_1^e x\,(\log x)^n \; dx$, show that $I_n = \frac{1}{2}e^2 - \frac{1}{2}nI_{n-1}$ for $n \geq 1$.

    b. Find I_0 and hence show that $I_4 = \frac{1}{4}(e^2 - 3)$.
29. Let $u_n = \int_0^{\frac{\pi}{2}} \cos^n x \; dx$.

    a. Prove that $u_n = \frac{n-1}{n} u_{n-2}$ for $n \geq 2$.

    b. Hence evaluate u_5
30. Find $\int \ln\left(\sqrt{x} + \sqrt{1+x}\right) dx$
31. Let $I_n = \int_0^1 (1-x^r)^n \, dx$, where $r > 0$, for $n = 0, 1, 2 \ldots$

a. Show that $I_n = \frac{nr}{nr+1} I_{n-1}$ for $n = 1, 2, 3 \ldots$

b. Hence evaluate $\int_0^1 \left(1 - x^{\frac{3}{2}}\right)^3 dx$

32. Let $I_n = \int_0^1 x(1-x)^n \, dx$ for $n = 0, 1, 2 \ldots$

a. Show that $I_n = \frac{n}{n+2} I_{n-1}$

b. Show that $I_n = \frac{1}{2({}^{n+2}C_2)}$

33. Given that $y = \cos^n x \sin nx$

a. Show that $\frac{dy}{dx} = 2n \cos^n x \cos nx - n \cos^{n-1} x \cos(n-1)x$. Hence show that $2n \int \cos^n x \cos nx \, dx - n \int \cos^{n-1} x \cos(n-1)x \, dx = \cos^n x \sin nx + C$

b. Using the result of part i, show that $\int_0^{\frac{\pi}{2}} \cos^n x \cos nx \, dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cos(n-1)x \, dx$

c. Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \cos nx \, dx$. Find I_1 and hence find I_8.

d. Hence find $\int_0^{\frac{\pi}{2}} \sin^8 x \cos 8x \, dx$

34. a. Show that $\int_{-a}^{a} f(x)dx = \int_{-a}^{a} f(-x)dx$

b. Hence or otherwise evaluate $\int_{-4}^{4} (e^x - e^{-x}) \cos x \, dx$

c. By considering $\int_0^a \sqrt{a^2 - x^2} dx$ find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

35. Evaluate $\int_0^{\frac{\pi}{4}} \frac{1 - \tan x}{1 + \tan x} dx$

36. a. Use the substitution $u = e^x$ to find $\int \frac{e^{2x}}{1+e^x} dx$

b. Find $\int \frac{\sin^3 x}{\cos^4 x} dx$

c. Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{4 + 5 \sin x} dx$

d. Evaluate $\int_0^1 \frac{\sin^{-1} x}{\sqrt{1+x}} dx$

37. a. Express $3 + 2x - x^2$ in the form $b^2 - (x-a)^2$. Hence evaluate $\int_1^3 \sqrt{3 + 2x - x^2} dx$

b. Find $\int x \sec^2 x \, dx$

c. Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x + \sin x} dx$. Hence use the substitution $u = \frac{\pi}{2} - x$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{x}{1 + \cos x + \sin x} dx$

38. a. Find $\int \frac{x+3}{(x+1)(x^2+1)} dx$

b. Use the substitution $x = 4\sin^2\theta$ to evaluate $\int_0^2 \sqrt{[x(4-x)]} \, dx$

c. i. If $I_n = \int_0^1 x^n e^x dx$ for $n \geq 0$ show that $I_n = e - nI_{n-1}$ for $n \geq 1$

ii. Find the value of I_4 and hence evaluate $\int_0^{\frac{1}{2}} x^4 e^{2x} dx$

Answer

1. $\frac{(\ln x)^2}{2}$
2. $\ln\left|\frac{(x-1)^2}{x+3}\right|$
3. $\ln\left|(x-2)\sqrt{x^2+1}\right|$
4. $48 + \ln 2$
5. $\frac{x}{4\sqrt{x^2+4}}$
6. $2\tan^{-1}\sqrt{x}$
7. $\frac{2}{3}(x+1)^{\frac{3}{2}} - 2(x+1)^{\frac{1}{2}}$
8. $\frac{\pi}{12}$
9. $\frac{2}{3}(\sin x)^{\frac{3}{2}} - \frac{2}{7}(\sin x)^{\frac{7}{2}}$
10. $-\frac{1}{10}\cos 5x - \frac{1}{6}\cos 3x$
11. $x^2e^x - 2xe^x + 2e^x$
12. $\frac{\pi}{4} - \frac{1}{2}\ln 2$
13. $2\ln 2 - \pi$
14. $\frac{1}{6}$
15. $\frac{\pi}{4} - \frac{3}{2}\ln 2$
16. proving
17. b. i. $1 - \frac{\pi}{4}$; ii. $\frac{\pi}{4}$
18. proving
19. 1; 2
20. $9e - 24$
21. 0
22. proving
23. $x\sec x - \ln(\sec x + \tan x) + C$
24. $\frac{1}{3}\sin 3\theta + \sin^3\theta + C$
25. proving
26. proving
27. proving
28. b. $\frac{1}{2}(e^2 - 1)$
29. b. $\frac{8}{15}$
30. $-\frac{1}{2}\sqrt{x}\sqrt{1+x} + \left(x + \frac{1}{2}\right)\ln\left(\sqrt{x} + \sqrt{1+x}\right) + c$
31. $\frac{81}{220}$
32. proving
33. c. $I_1 = \frac{\pi}{4}, I_8 = \frac{\pi}{512}$; d. $\frac{\pi}{512}$
34. b. $I = 0$; c. πab
35. $\ln 2$
36. a. $e^x - \ln(e^x + 1)$; b. $\frac{1}{3}\sec^3 x - \sec x$; c. $\frac{1}{3}\ln 2$; d. $\pi\sqrt{2} - 4$
37. a. $b = 2, a = 1, \pi$; b. $x\tan x + \ln|\cos x|$; c. $\ln 2, \frac{\pi}{4}\ln 2$
38. a. $\ln\left|\frac{x+1}{\sqrt{x^2+1}}\right| + 2\tan^{-1}x$; b. π; c. $9e - 24, \frac{1}{32}(9e - 24)$